

COMPANION TO THE REPORT “ENCODING THE JERSEY COVID-19 GATHERING CONTROL ORDER IN L4”

Reading L4: the Gathering Control Order

Annotated excerpts from the three-version GCO encoding

Legalese · prepared for CALC 2026, Singapore · 16 July 2026¹

The report tells the story of encoding the *Covid-19 (Gathering Control) (Jersey) Order 2020* — a small, fast-moving pandemic instrument — three times over: as first enacted, as at 15 January 2021, and as last in force before repeal. This companion shows the encoding itself. Where the Charities booklet taught the language’s core, these six excerpts show what a subordinate instrument under weekly amendment demands: *computed facts*, *numeric thresholds*, *named exception gates*, *typed holes for open texture*, an *exported API*, and above all *amendment honesty* — three parallel files whose disagreements are executable. Sidebars cite the report’s own sections.

Excerpt 1 — The Interpretation article, made tabular

Article 1 habits become the domain model. Note the fields that are not *asked* but *computed*.

GCO-as-repealed.l4, lines 9–39 (abridged)

```

DECLARE VenueKind IS ONE OF
  Indoor
  Outdoor

DECLARE Gathering HAS
  `number of persons taking part`           IS A NUMBER
  `takes place indoors`                     IS A BOOLEAN
  `takes place outdoors`                    IS A BOOLEAN
  MEANS NOT `takes place indoors`
  `venue kind`                               IS A VenueKind
  MEANS IF `takes place indoors` THEN Indoor ELSE Outdoor
  `is a school group of pupils and staff at school premises` IS A BOOLEAN
  `is a wedding, civil partnership ceremony or funeral` IS A BOOLEAN
  `number of ceremony officers`              IS A NUMBER
  `number of guests`                         IS A NUMBER
  `total of officers and guests`              IS A NUMBER
  MEANS `number of ceremony officers` PLUS `number of guests`
  `is for the purposes of work`              IS A BOOLEAN

DECLARE PubliclyAccessiblePremisesGathering HAS
  `number of persons gathering in the indoor areas` IS A NUMBER
  `premises are publicly accessible`               IS A BOOLEAN
  `the person is responsible for the premises`      IS A BOOLEAN
  `the person allowed the gathering`                IS A BOOLEAN

```

Sidebar 1 · Computed fields — MEANS inside a DECLARE

`takes place indoors` is the one authoritative input; `takes place outdoors` **MEANS NOT** it, and `venue kind` folds the boolean into a two-valued kind. Nobody can ever assert a gathering that is both indoors and outdoors, because the derived facts cannot be set, only computed — the report calls two independent booleans “a silent bug farm” (§3.3). Likewise `total of officers and guests` is a single derived number (**PLUS**), so Article 5’s cap attaches to one well-defined quantity. This is the Interpretation article doing what it always did — fixing the vocabulary — with the arithmetic now carried by the definitions themselves (report, Step 1: “the **DECLARE** block is not a software invention; it is the Interpretation article made tabular”).

Sidebar 2 · Two records, not one “Event” with optional everything

Article 3’s world (headcount, venue, exception flags) and Article 7’s world (indoor headcount on publicly accessible premises, responsibility, allowing) are kept as *separate* records, mirroring the Order’s own split. The alternative — one giant type with many optional fields — is how encodings become unreadable: “optional soup” (report, Step 2). The two situation records for the offence questions each pair one gathering with one actor.

¹Prepared with AI drafting assistance; excerpts are verbatim from the corpus files with locations cited; elisions are marked [. . .]. A first booklet in this series annotates the Charities (Jersey) Law 2014 encoding; its ten sidebars cover the core syntax (**DECLARE**/**HAS**, **GIVEN**/**GIVETH**, backtick names, **WITH**, **#ASSERT**) and are assumed rather than repeated here.

Excerpt 2 — One Article, three points in time

The same two thresholds, from the three parallel files. Watch the numbers — and watch what is being *counted*.

GCO-first-version.l4, lines 87–92 and 108–112 (in force 4–14 Dec 2020)

```

§§ `3 Restriction on gatherings - as first enacted`
-- As first enacted Art.3 applied to "any premises" without indoor/outdoor
-- distinction: "A person must not take part in a gathering of more than 10
-- people at any premises."

GIVEN g IS A Gathering
GIVETH A BOOLEAN
DECIDE `Article 3 size limit is exceeded` g IF
  g's `number of persons taking part` GREATER THAN 10

DECIDE `Article 5 size cap is respected` g IF
  g's `is a wedding, civil partnership ceremony or funeral`
  AND g's `total of officers and guests` AT MOST 20

```

GCO-as-at-20210115.l4, lines 99–116 (abridged; mid-chain consolidation)

```

§§ `5 Exception - wedding, civil partnership or funeral`
-- DIFF vs first enacted: exception may overshoot the Art.3 outdoor/indoor
-- roofs (not a flat "more than 10").
-- DIFF vs last in force (post R&O.11/2021 "20 people in total"):
-- Mid-chain wording after R&O.1/2021 (and until R&O.11 rewrote it) capped the
-- GUEST count at 20; ceremony officers sat outside that guest roof.

DECIDE `Article 5 size cap is respected` g IF
  g's `is a wedding, civil partnership ceremony or funeral`
  AND g's `number of guests` AT MOST 20

```

GCO-as-repealed.l4, lines 61–83 and 107–123 (abridged; as last in force)

```

§§ `3 Restriction on gatherings - as last in force`
-- DIFF: first enacted flat 10-person cap at any premises.
-- Last in force splits venue kind (amended by R&O.189/2020 and L.6/2021):
-- Art.3(1) outdoor gathering of more than 20 people
-- Art.3(2) indoor gathering of more than 10 people

DECIDE `Article 3 outdoor size limit is exceeded` g IF
  g's `takes place outdoors`
  AND g's `number of persons taking part` GREATER THAN 20

DECIDE `Article 3 indoor size limit is exceeded` g IF
  g's `takes place indoors`
  AND g's `number of persons taking part` GREATER THAN 10

DECIDE `Article 5 outdoor size cap is respected` g IF
  g's `takes place outdoors`
  AND g's `total of officers and guests` AT MOST 20
-- [...] indoor twin kept separate, "so a later re-tightening of either limb
-- is easy to re-encode"

```

Sidebar 3 · DECIDE ... IF — the light idiom, and numbers

This corpus reads differently from the Charities one: decisions are **DECIDE** *name* **g IF** ..., without the inline verbatim statute strings. Two idioms, one language: the heavyweight quotation style earns its keep on a 43-Article Law; for a ten-page Order the prose-length backtick names *are* the isomorphism — “a legal reader diffs against Art.3 without a legend” (report §3.6). The numeric vocabulary is deliberately plain English: **GREATER THAN**, **AT MOST**, **PLUS**, **EQUALS**.

Sidebar 4 · Amendment honesty — parallel files with – DIFF: markers

Three isomorphic skeletons, one per point in time, with – DIFF: markers where the regime moved. Drafters review by comparing *versions of the regime*, not commits (report, Step 6). And the diffs are subtler than “10 became 20”: Article 5’s cap changed *what it counts* — officers + guests in total (first enacted), then guests only with officers outside the roof (R&O.1/2021), then back to a 20-person total (R&O.11/2021). A single “current” file with narrative

comments would bury exactly the kind of change a drafting office most needs to see.

Excerpt 3 — Exceptions as named gates

Articles 4–6 each become one named predicate; Article 3 then reads as the statute does: size exceeded, minus the carve-outs.

GCO-as-repealed.l4, lines 85–157 (abridged)

```

DECIDE `Article 4 school-group exception applies` g IF
  g's `is a school group of pupils and staff at school premises`

DECIDE `Article 5 wedding funeral exception applies` g IF
  `Article 5 wedding funeral exception is available` g
  AND `Article 5 size cap is respected` g

DECIDE `Article 6 work exception applies` g IF
  g's `is for the purposes of work`

§§ `3 read with 4, 5 and 6 - restricted gathering`

DECIDE `an Article 3 exception applies` g IF
  `Article 4 school-group exception applies` g
  OR `Article 5 wedding funeral exception applies` g
  OR `Article 6 work exception applies` g

DECIDE `the gathering is a restricted gathering under Article 3` g IF
  `Article 3 size limit is exceeded` g
  AND NOT `an Article 3 exception applies` g

DECIDE `a person must not take part in the gathering under Article 3` g IF
  `the gathering is a restricted gathering under Article 3` g

```

Sidebar 5 · Named gates beat buried ORs

The whole regime in one line: $restricted \Leftrightarrow size\ exceeded \wedge \neg(Art.4 \vee Art.5 \vee Art.6)$. The report calls naming the exception gates “the single highest return-for-effort improvement for audit” (Step 3): each gate carries its Article number in its name, so a reviewer line-matches the Order clause by clause, and the **AND NOT** defeater shape — the same shape as the Charities knock-out — has nowhere to hide an extra condition. A buried mega-**OR** looks clever and fails that review the first time someone tries it.

Excerpt 4 — Open texture as typed holes

Two residual defences deliberately point *outside* the Order: to ministerial guidance, and to one of the most specific factual classes in the statute book.

GCO-as-repealed.l4, lines 41–51

```

DECLARE Participant HAS
  `name` IS A STRING
  `is a critical worker as defined in guidance published by the Minister for Health and Social Services` IS A
    BOOLEAN
  `fails to take part by reason of force majeure as a critical worker` IS A BOOLEAN
    MEANS `is a critical worker as defined in guidance published by the Minister for Health and Social
    Services`

DECLARE ResponsiblePerson HAS
  `name` IS A STRING
  `is an official from Visit Jersey escorting a pre-arranged automotive tour of the island for news and travel
  journalists who are guests of the Jersey Government` IS A BOOLEAN
  `fails to prevent the gathering by reason of force majeure as a holiday industrial tour guide` IS A BOOLEAN
    MEANS `is an official from Visit Jersey escorting a pre-arranged automotive tour of the island for news
    and travel journalists who are guests of the Jersey Government`

```

Sidebar 6 · The honest answer to “how do you encode open texture?”

You don't. Critical-worker status is defined by guidance published by the Minister — it cannot be deduced from the Order by any amount of logic, so the encoding types the residual as a **BOOLEAN** and forces every caller — human form, test case, or agent API — to assert it. “Evaluation never *guesses*” (report, Step 4). The Visit Jersey tour-escort defence makes the same point from the other end: a factual class so concrete it is almost a joke, and still a fact the model must be *told*. Any claim that an encoding “decides GCO compliance end-to-end without human residual input” is false for this Order, and the report says so out loud (§3.7). This is the measured penumbra again: the model does not shrink the grey area; it shows you exactly where it is and bills it to the right decision-maker.

Excerpt 5 — Offence, fine, and the exported API

The criminal layer is three short definitions — and the `@export` annotations that turn the file into a callable service.

GCO-as-repealed.l4, lines 174–195 (abridged)

```

§§ `9 Offences and penalties`

@export Whether the participant commits an Article 3 offence (taking part in a restricted gathering, subject to
  ↳ the critical-worker force-majeure defence)
GIVEN s IS A Article3Situation @desc Situation under Art.3: the gathering facts plus the participant's
  ↳ critical-worker status
GIVETH A BOOLEAN
DECIDE `the person commits an Article 3 offence` IF
  `the gathering is a restricted gathering under Article 3` (s's gathering)
  AND NOT s's participant's `fails to take part by reason of force majeure as a critical worker`

@export Whether the responsible person commits an Article 7 offence [...]
DECIDE `the person commits an Article 7 offence` IF
  `the responsible person must not allow the gathering under Article 7` (s's gathering)
  AND s's gathering's `the person allowed the gathering`
  AND NOT s's responsible's `fails to prevent the gathering by reason of force majeure as a holiday industrial
  ↳ tour guide`

@export The maximum fine in pounds for an Article 3 or Article 7 offence under this Order (level 2 on the
  ↳ standard scale: 1000)
GIVEN has_committed_offence IS A BOOLEAN @desc TRUE if the person has committed an Art.3 or Art.7 offence
GIVETH A NUMBER
DECIDE `Article 9 maximum fine in pounds` IS
  IF has_committed_offence THEN 1000 ELSE 0

```

Sidebar 7 · Offence = prohibition $\wedge$ act $\wedge$ $\neg$ defence — kept light

Criminal “commits if” is a *constitutive* pattern layered on a prohibitive fact pattern, so that is all the machinery used: no **PARTY** ... **MUST** ... **HENCE/LEST** process model anywhere in this corpus. The Charities encoding needed the full deontic layer because its source chains duties through time; the GCO’s “must not take part” is a *duty-fact*, and ramming it into a process model “would have been tool-worship” (report §3.5). Matching the weight of the formal machinery to the weight of the source is itself a modelling skill. Note also the chained possessive: *s’s gathering’s* ‘the person allowed the gathering’ reads a fact two records deep.

Sidebar 8 · @export / @desc — the same file is the API

Five entry points carry `@export`, each **GIVEN** parameter a `@desc` written for a consumer with zero project context. That is what a web form, an OpenAPI schema, or an MCP tool call binds to — the guardrailed reasoner the LLM consults instead of improvising. “Publishing is different from encoding. An encoding nobody can call from a form or an agent remains a research artefact” (report, Step 8).

Excerpt 6 — Golden cases, and the flip the amendment caused

Every file ends with the same named gatherings and the checks they must pass. One scenario — eleven people in a field — changes its legal answer across the amendment chain, and the two files *assert opposite things*.

GCO-as-repealed.l4, lines 221–230 and 340–358 (abridged), with one line from GCO-first-version.l4

```
-- DIFF: outdoor of 11 was prohibited under first enacted (flat 10); lawful
-- under last in force (outdoor rooftop 20)
`ordinary outdoor of 11` MEANS Gathering WITH
  `number of persons taking part`           IS 11
  `takes place indoors`                     IS FALSE
-- [...] no school / ceremony / work flags

-- GCO-first-version.l4, line 296 (in force 4-14 Dec 2020):
#ASSERT `a person must not take part under Article 3` `ordinary outdoor of 11`

-- GCO-as-repealed.l4, line 341 (as last in force):
#ASSERT NOT (`a person must not take part under Article 3` `ordinary outdoor of 11`)

-- and the rest of the golden set, as last in force:
#ASSERT `a person must not take part under Article 3` `ordinary indoor of 11`
#ASSERT NOT (`a person must not take part under Article 3` `school of 30`)
#ASSERT NOT (`a person must not take part under Article 3` `wedding of 2 officers and 18 guests`)
#ASSERT `a person must not take part under Article 3` `wedding of 2 officers and 19 guests over cap`
#ASSERT NOT (`a person must not take part under Article 3` `worksite of 25`)
#ASSERT NOT (`the person commits an Article 3 offence` `Chris critical worker at ordinary indoor of 11`)
#ASSERT NOT (`the person commits an Article 7 offence` `Vic tour guide allows pub indoor of 12`)
#ASSERT `Article 9 maximum fine in pounds` TRUE EQUALS 1000
```

Sidebar 9 · The executable disagreement

The first-enacted file asserts the outdoor gathering of eleven is prohibited; the as-repealed file asserts it is not. Both assertions are true — of different laws — and keeping both as living, machine-checked files “makes the disagreement executable rather than narrative” (report §3.4). The golden set is the specificity of the encoding’s claim: school group saved, wedding saved at the cap and lost one guest over it, critical worker excused, tour guide excused. “Without them, ‘we encoded Art.3’ is marketing” (report, Step 7). For a drafting office, this is the pre-mortem made permanent: every scenario anyone ever worried about stays checked, on every future amendment, at zero marginal cost.

Corpus: GCO/ — GCO-first-version.l4, GCO-as-at-20210115.l4, GCO-as-repealed.l4, the 10-page report, and a drafter-facing README. Run `14 check / 14 run` on any of the three files to re-verify the golden cases. A fourth file, MHO-as-at-20210115.l4, encodes the Covid-19 (Mental Health) Order: the `jerseylaw.je` point-in-time slug for 15 January 2021 serves *that* instrument in place of the Gathering Control Order, so even finding the right consolidated text is part of the craft. Excerpt line numbers refer to the files as of 14 July 2026.